

CS 237: PROBABILITY IN COMPUTING

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MODIFIED BAYES' THEOREM:

$$P(H|X) = P(H) * \left(1 + P(C) * \left(\frac{P(x|H)}{P(x)} - 1 \right) \right)$$

H: HYPOTHESIS

x: OBSERVATION

P(H): PRIOR PROBABILITY THAT H IS TRUE

P(x): PRIOR PROBABILITY OF OBSERVING x

P(C): PROBABILITY THAT YOU'RE USING
BAYESIAN STATISTICS CORRECTLY

LECTURE 11

Last time

- Quiz 2
- Law of Total Probability
- Bayes Rule

Today

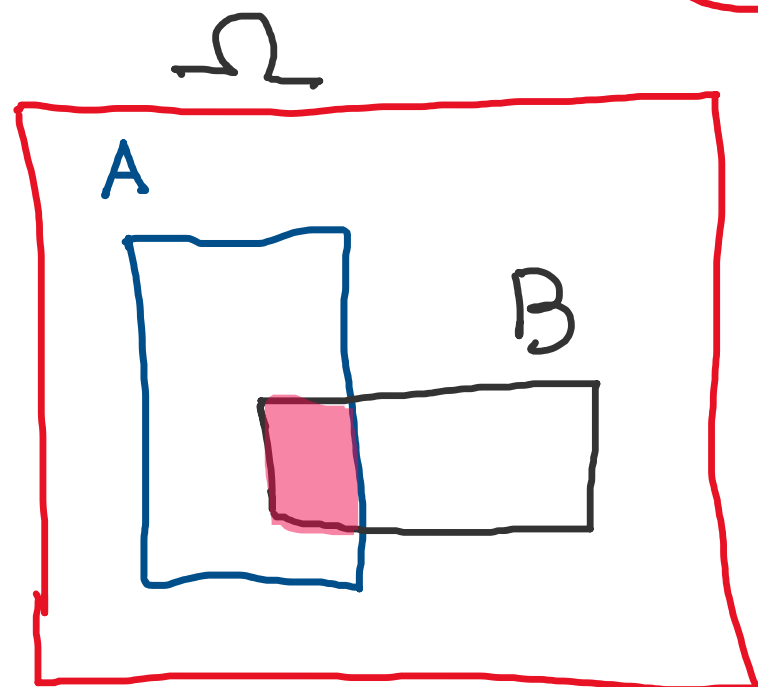
- Independence

<https://xkcd.com/2059/>

INDEPENDENT EVENTS

- ▶ The conditional probability $\mathbb{P}(A \mid B)$ captures the partial information that event B provides about event A
- ▶ An important setting is when the information that B happened does not change the probability that A happened:

$$\mathbb{P}(A \mid B) = \mathbb{P}(A) \Rightarrow \mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$$



$$Pr(A \cap B) = Pr(A|B) \cdot Pr(B)$$

$$= Pr(A) \cdot Pr(B)$$

INDEPENDENT EVENTS

- ▶ **Definition:** Two events A and B are **independent** if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$$

- ▶ Observe: $\mathbb{P}_r(A|B) = \mathbb{P}_r(A)$ and $\mathbb{P}_r(B|A) = \mathbb{P}_r(B)$
 - ▶ Independence is **symmetric**: if A is independent of B then B is independent of A
 - ▶ The above definition can be used even when $\mathbb{P}(B) = 0$
 - ▶ An event B with $\mathbb{P}(B) = 0$ is independent of all events, including itself

ASK THE AUDIENCE

- ▶ **Experiment:** toss a coin twice, all outcomes equally likely
- ▶ $A = \text{"first toss is heads"}$ $\Pr(A) = \frac{1}{2}$
- ▶ $B = \text{"second toss is heads"}$ $\Pr(B) = \frac{1}{2}$

$$\Pr(A \cap B) = \Pr(HH) = \frac{1}{4} = \Pr(A) \cdot \Pr(B)$$

Claim: A and B independent

(A) True

(B) False

ASK THE AUDIENCE

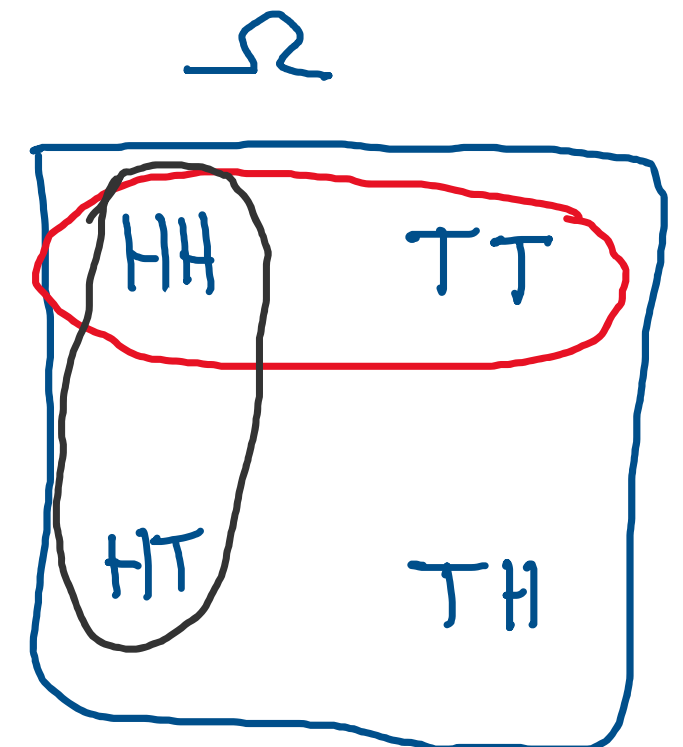
- ▶ **Experiment:** toss two (fair, independent) coins
- ▶ $A = \text{"first coin toss is heads"}$ $\Pr(A) = \frac{1}{2}$
- ▶ $C = \text{"tosses match (both H or both T)"}$ $\Pr(C) = \Pr(\{HH, TT\}) = \frac{1}{2}$

$$\Pr(A \cap C) = \Pr(HH) = \frac{1}{4} = \Pr(A) \cdot \Pr(C)$$

Claim: A and C are independent

(A) True

(B) False



ASK THE AUDIENCE

- ▶ **Experiment:** toss a **biased** coin with $\mathbb{P}(H) = 0.7$ **twice**
- ▶ $A = \text{"first toss is heads"}$
- ▶ $C = \text{"tosses match (both H or both T)"}$

Are A and C independent?

(A) True

(B) False

ASK THE AUDIENCE

- ▶ **Experiment:** toss a **biased** coin with $\mathbb{P}(H) = 0.7$ **twice**
- ▶ $A = \text{"first toss is heads"}$ $\Pr(\{HH, HT\}) = 0.49 + 0.21 = 0.7$
- ▶ $C = \text{"tosses match (both H or both T)"}$ $= \Pr(\{HH, TT\}) = \Pr(HH) + \Pr(TT)$
 $= 0.49 + 0.09 = 0.58$

$$\Pr(A \cap C) = \Pr(\{HH\}) = 0.49 \neq 0.7 \times 0.58 = 0.406$$

A and C are Not independent

ASK THE AUDIENCE

- True or false: If A and B are disjoint, they are independent

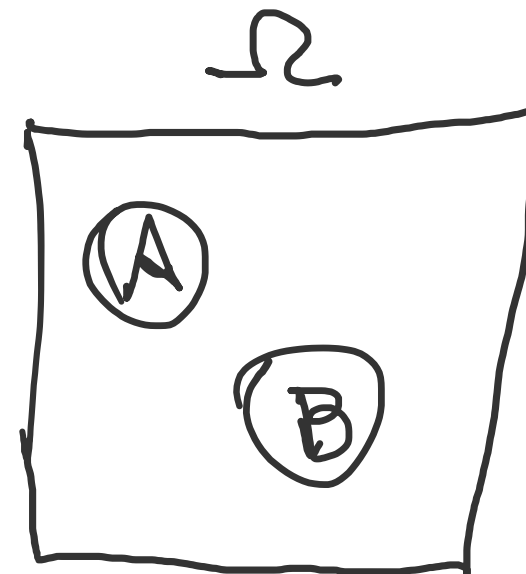
$$P_r(A \cap B) = 0 \neq P_r(A) \cdot P_r(B)$$

(A) True

(B) False

$$P_r(A) > 0$$

$$P_r(B) > 0$$



INDEPENDENCE IS NOT ALWAYS INTUITIVE

- ▶ Experiment: roll a 4-sided die twice (independently)
- ▶ A = "first roll is 1" $P_r(A) = \frac{1}{4}$
- ▶ B = "sum of the two rolls is 5" $P_r(B) = P_r(\{(1,4), (2,3), (3,2), (4,1)\}) = \frac{1}{4}$

$$P_r(A \cap B) = P_r(\{(1,4)\}) = \frac{1}{16} = P_r(A) \cdot P_r(B) = \frac{1}{4} \cdot \frac{1}{4}$$

Are A and B independent?

(A) Yes

(B) No

INDEPENDENCE IS NOT ALWAYS INTUITIVE

- ▶ Experiment: roll a 6-sided die twice (independently)
- ▶ $A = \text{"first roll is 1"} \quad \Pr(A) = \frac{1}{6}$
- ▶ $B = \text{"sum of the two rolls is 5"} \quad \Pr(B) = \Pr(\{(1,4), (2,3), (3,2), (4,1)\}) = \frac{4}{36}$

$$\Pr(A \cap B) = \Pr(\{(1,4)\}) = \frac{1}{36} \neq \frac{1}{6} \cdot \frac{4}{36} = \frac{1}{54}$$

Are A and B independent?

(A) Yes

(B) No

INDEPENDENCE FOR 3 OR MORE EVENTS

- ▶ Recall that two events A and B are **independent** if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$$

- ▶ Our next task is to define independence for more than two events

INDEPENDENCE FOR 3 EVENTS

- ▶ What is the right definition of independence for 3 events A, B, C ?

1. Pairwise independence:

- ▶ A, B independent: $\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$
- ▶ A, C independent: $\mathbb{P}(A \cap C) = \mathbb{P}(A) \cdot \mathbb{P}(C)$
- ▶ B, C independent: $\mathbb{P}(B \cap C) = \mathbb{P}(B) \cdot \mathbb{P}(C)$

2. Triple equality: $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A) \cdot \mathbb{P}(B) \cdot \mathbb{P}(C)$

3. Require both: require both 1 and 2

INDEPENDENCE FOR 3 EVENTS

1. Pairwise independence:

- ▶ A, B independent: $\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$
- ▶ A, C independent: $\mathbb{P}(A \cap C) = \mathbb{P}(A) \cdot \mathbb{P}(C)$
- ▶ B, C independent: $\mathbb{P}(B \cap C) = \mathbb{P}(B) \cdot \mathbb{P}(C)$

2. Triple equality : $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A) \cdot \mathbb{P}(B) \cdot \mathbb{P}(C)$

3. Questions: does 1 imply 2? does 2 imply 1?

INDEPENDENCE FOR 3 EVENTS

- Pairwise independence does **not** imply the triple equality
- Experiment:** toss a coin 2 times (independently)

- $A = \text{"first toss H"} = \{HH, HT\}$

- $B = \text{"second toss H"} = \{HH, TH\}$

- $C = \text{"tosses are different"} = \{HT, TH\}$

*A and B
are independent*

*A and C
are independent*

B and C are independent

$$\Pr(A \cap B \cap C) = \Pr(A) \cdot \Pr(B) \cdot \Pr(C)$$

$$\Pr(\emptyset) = 0 \neq \left(\frac{1}{2}\right)^3$$

INDEPENDENCE FOR 3 EVENTS

- ▶ The triple equality does **not** imply pairwise independence
- ▶ **Experiment:** roll a 6-sided die twice (independently)
- ▶ $A = \text{"first roll is 1, 2, or 3"} \quad \Pr(A) = \frac{1}{2}$
- ▶ $B = \text{"first roll is 3, 4, or 5"} \quad \Pr(B) = \frac{1}{2}$
- ▶ $C = \text{"the sum of the two rolls is 9"} \quad \Pr(C) = \Pr(\{(3,6), (4,5), (5,4), (6,3)\}) = \frac{4}{36}$

$$\Pr(A \cap B \cap C) = \Pr(\{(3,6)\}) = \frac{1}{36} = \Pr(A) \cdot \Pr(B) \cdot \Pr(C) = \left(\frac{1}{2}\right)^2 \cdot \frac{4}{36}$$

Homework: check pairwise independence

EXAMPLE

- ▶ Experiment: toss a coin 2 times (independently)
- ▶ A_1 = "first toss is H"
- ▶ A_2 = "second toss is H"
- ▶ A_3 = "the two tosses are different"

As we just noticed:

A_1, A_2, A_3 are pairwise independent

A_1, A_2, A_3 are not mutually independent

INDEPENDENCE

- ▶ Independence is a very strong assumption
- ▶ Many incorrect arguments in probability are due to incorrect assumptions on the independence of events



MUTUAL INDEPENDENCE

- ▶ We can generalize the definition to any number of events
- ▶ **Definition** (mutual independence for n events):
The events $A_1, A_2, \dots, A_n$ are mutually independent if, for **every subset** $S \subseteq \{1, 2, \dots, n\}$, we have

$$\mathbb{P}\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} \mathbb{P}(A_i)$$

For every subset S , the probability that all of the events in S happen is equal to the product of the probabilities

$$\sum_{i=1}^n A_i = A_1 + A_2 + \dots + A_n \quad (\text{numbers})$$

$$\bigcap_{i=1}^n A_i = A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n \quad (\text{sets})$$

$$\prod_{i=1}^n A_i = A_1 \cdot A_2 \cdot A_3 \cdot \dots \cdot A_n \quad (\text{numbers})$$

MUTUAL INDEPENDENCE

- ▶ Example: for 4 events, we require **all** of the 11 conditions:

$$\mathbb{P}(A_1 \cap A_2) = \mathbb{P}(A_1)\mathbb{P}(A_2)$$

$$\mathbb{P}(A_1 \cap A_3) = \mathbb{P}(A_1)\mathbb{P}(A_3)$$

$$\mathbb{P}(A_1 \cap A_4) = \mathbb{P}(A_1)\mathbb{P}(A_4)$$

$$\mathbb{P}(A_2 \cap A_3) = \mathbb{P}(A_2)\mathbb{P}(A_3)$$

$$\mathbb{P}(A_2 \cap A_4) = \mathbb{P}(A_2)\mathbb{P}(A_4)$$

$$\mathbb{P}(A_3 \cap A_4) = \mathbb{P}(A_3)\mathbb{P}(A_4)$$

$$\mathbb{P}(A_1 \cap A_2 \cap A_3) = \mathbb{P}(A_1)\mathbb{P}(A_2)\mathbb{P}(A_3)$$

$$\mathbb{P}(A_1 \cap A_2 \cap A_4) = \mathbb{P}(A_1)\mathbb{P}(A_2)\mathbb{P}(A_4)$$

$$\mathbb{P}(A_1 \cap A_3 \cap A_4) = \mathbb{P}(A_1)\mathbb{P}(A_3)\mathbb{P}(A_4)$$

$$\mathbb{P}(A_2 \cap A_3 \cap A_4) = \mathbb{P}(A_2)\mathbb{P}(A_3)\mathbb{P}(A_4)$$

$$\mathbb{P}(A_1 \cap A_2 \cap A_3 \cap A_4) = \mathbb{P}(A_1)\mathbb{P}(A_2)\mathbb{P}(A_3)\mathbb{P}(A_4)$$

MUTUAL INDEPENDENCE

- ▶ Example: 4 events A_1, A_2, A_3, A_4

$$\mathbb{P}(A_1 \cap A_2) = \mathbb{P}(A_1)\mathbb{P}(A_2)$$

$$\mathbb{P}(A_1 \cap A_3) = \mathbb{P}(A_1)\mathbb{P}(A_3)$$

$$\mathbb{P}(A_1 \cap A_4) = \mathbb{P}(A_1)\mathbb{P}(A_4)$$

$$\mathbb{P}(A_2 \cap A_3) = \mathbb{P}(A_2)\mathbb{P}(A_3)$$

$$\mathbb{P}(A_2 \cap A_4) = \mathbb{P}(A_2)\mathbb{P}(A_4)$$

$$\mathbb{P}(A_3 \cap A_4) = \mathbb{P}(A_3)\mathbb{P}(A_4)$$

 $\Rightarrow$

pairwise independent
(or 2-wise independent)

MUTUAL INDEPENDENCE

- ▶ Example: 4 events A_1, A_2, A_3, A_4

pairwise independent

+

$$\mathbb{P}(A_1 \cap A_2 \cap A_3) = \mathbb{P}(A_1)\mathbb{P}(A_2)\mathbb{P}(A_3)$$

$$\mathbb{P}(A_1 \cap A_2 \cap A_4) = \mathbb{P}(A_1)\mathbb{P}(A_2)\mathbb{P}(A_4)$$

$$\mathbb{P}(A_1 \cap A_3 \cap A_4) = \mathbb{P}(A_1)\mathbb{P}(A_3)\mathbb{P}(A_4)$$

$$\mathbb{P}(A_2 \cap A_3 \cap A_4) = \mathbb{P}(A_2)\mathbb{P}(A_3)\mathbb{P}(A_4)$$

$\Rightarrow$ 3-wise independent

MUTUAL INDEPENDENCE

- ▶ Example: 4 events A_1, A_2, A_3, A_4

3-wise independent

+

$$\mathbb{P}(A_1 \cap A_2 \cap A_3 \cap A_4) = \mathbb{P}(A_1)\mathbb{P}(A_2)\mathbb{P}(A_3)\mathbb{P}(A_4)$$

$\Downarrow$

4-wise independent

(mutually independent)

k -WISE INDEPENDENCE

- ▶ **Definition** (k -wise independence for n events):
The events $A_1, A_2, \dots, A_n$ are k -wise independent if, for **every subset** $S \subseteq \{1, 2, \dots, n\}$ with $|S| \leq k$ we have

$$\mathbb{P}\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} \mathbb{P}(A_i)$$

So, mutual independence for n events is exactly n -wise independence.

RECAP

Definition of independence:

$$\Pr(A \cap B) = \Pr(A) \cdot \Pr(B)$$

also $\Pr(A|B) = \Pr(A)$ and $\Pr(B|A) = \Pr(B)$
when $\Pr(A) > 0$, $\Pr(B) > 0$

PEOPLE V. COLLINS



From <http://www.stus.com/stus-cartoon.php?name=People+v.+Collins&cartoon=evi0007>

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- ▶ Robbery case in 1968 in a suburb of LA
- ▶ A purse was snatched from an elderly person
- ▶ A couple seen running from the scene were described as a black man with a beard and a mustache and a blond girl with hair in a ponytail
- ▶ Witnesses said they drove off in a partly yellow car

https://en.wikipedia.org/wiki/People_v._Collins

PEOPLE V. COLLINS

- ▶ Malcolm and Janet Collins were arrested
- ▶ He was black and, though clean shaven when arrested, had evidence of recently having had a beard and a mustache
- ▶ She was blond and usually wore her hair in a ponytail
- ▶ They drove a partly yellow Lincoln

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- ▶ The prosecution called a mathematics instructor as a witness, who testified about the multiplication rule for Pr
- ▶ The prosecutor suggested to the jury to consider the following odds that the defenders were not the robbers

Black man with beard	1 in 10
Man with mustache	1 in 4
White woman with pony tail	1 in 10
White woman with blond hair	1 in 3
Yellow motor car	1 in 10
Interracial couple in car	1 in 1,000

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- ▶ Using the product rule, the prosecutor argued that the probability the Collin's were innocent was

$$\frac{1}{10} \cdot \frac{1}{4} \cdot \frac{1}{10} \cdot \frac{1}{3} \cdot \frac{1}{10} \cdot \frac{1}{1,000} = \frac{1}{12,000,000}$$

ASK THE AUDIENCE

What do you think the jury ruled?

(A) Guilty

(B) Not guilty

ASK THE AUDIENCE

- ▶ The jury ruled guilty

Do you agree with their decision?

(A) Yes

(B) No

ASK THE AUDIENCE

- ▶ The jury ruled guilty

Do you agree with their decision?

(A) Yes

(B) No

- ▶ Some stats:

$$\frac{1}{10} \cdot \frac{1}{4} \cdot \frac{1}{10} \cdot \frac{1}{3} \cdot \frac{1}{10} \cdot \frac{1}{1000} = \frac{1}{12,000,000}$$

Let's estimate there were 5,000,000 couples in the LA region

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- ▶ Let n be the number of couples in the LA suburb
- ▶ Modeling assumption: each couple matches the witnesses' description with probability p , independently of the other couples

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- ▶ Let n be the number of couples in the LA suburb
- ▶ Modeling assumption: each couple matches the witnesses' description with probability p , independently of the other couples
- ▶ Prosecutor's argument:

$$\begin{aligned} p &= \mathbb{P}(\text{"black man with beard"}) \mathbb{P}(\text{"man with mustache"}) \dots \\ &= \frac{1}{10} \cdot \frac{1}{4} \cdot \frac{1}{10} \cdot \frac{1}{3} \cdot \frac{1}{10} \cdot \frac{1}{1000} = \frac{1}{12,000,000} \end{aligned}$$

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$$\mathbb{P}(\text{"accused couple are not the robbers"}) = p$$

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- ▶ Prosecutor's argument:

$$\begin{aligned} p &= \mathbb{P}(\text{"black man with beard"}) \mathbb{P}(\text{"man with mustache"})... \\ &= \frac{1}{10} \cdot \frac{1}{4} \cdot \frac{1}{10} \cdot \frac{1}{3} \cdot \frac{1}{10} \cdot \frac{1}{1000} = \frac{1}{12,000,000} \end{aligned}$$

Assumes those events are mutually independent

Is it reasonable to assume, e.g., that "black man with beard" and "man with mustache" are independent events?

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- ▶ Prosecutor's argument:

$$\begin{aligned} &\mathbb{P}(\text{"accused couple are not the robbers"}) \\ &= \mathbb{P}(\text{"a random couple matches the description"}) \end{aligned}$$

Prosecutor's Fallacy

"The odds of finding this evidence on an innocent man are so small that the jury can safely disregard the possibility that this defendant is innocent"

https://en.wikipedia.org/wiki/Prosecutor's_fallacy

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- ▶ Prosecutor's argument:

$$\begin{aligned} & \mathbb{P}(\text{"accused couple are not the robbers"}) \\ &= \mathbb{P}(\text{"a random couple matches the description"}) \end{aligned}$$

Prosecutor's Fallacy

The odds of winning the lottery are so small that we can safely assume that the lottery winner cheated

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Prosecutor's Fallacy

The odds of winning the lottery are so small that we can safely assume that the lottery winner cheated

Say Tiago won the lottery. Now a prosecutor says he cheated:

$$\begin{aligned} & \mathbb{P}(\text{"Tiago won fairly (is not a cheater)"}) \\ &= \mathbb{P}(\text{"a random person wins the lottery"}) = 1 \text{ in a million} \end{aligned}$$

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- ▶ Prosecutor's argument:

$$\begin{aligned} &\mathbb{P}(\text{"accused couple are not the robbers"}) \\ &= \mathbb{P}(\text{"a random couple matches the description"}) \end{aligned}$$

To cast reasonable doubt, what do we need to compute?

$\mathbb{P}(\text{"there are at least 2 couples matching the description"})$

This is **just like** the medical testing examples. *Even if the test is very accurate, it does not mean you have the disease.*

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- ▶ Prosecutor's argument:

$$\begin{aligned} &\mathbb{P}(\text{"accused couple are not the robbers"}) \\ &= \mathbb{P}(\text{"a random couple matches the description"}) \end{aligned}$$

Even more, we need to consider the **conditional** probability

$$\begin{aligned} &\mathbb{P}(\text{"there are at least 2 couples matching description"} \\ &\quad | \text{"there is at least 1 couple matching description"}) \end{aligned}$$

because we know that there is at least couple in existence matching the description.

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- ▶ Suppose for example that:
 - ▶ There are $n = 5,000,000$ couples in the LA suburb
 - ▶ Each couple matches the witnesses' description with probability $p = 1/12,000,000$, independently of the others
- ▶ Find and compare the following probabilities:

$\mathbb{P}(\text{"at least two such couples"})$

$\mathbb{P}(\text{"at least two such couples"} \mid \text{"at least one such couple"})$

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- ▶ n couples, each matching description with probability p , independently of the other couples

- ▶ Example: $n = 5,000,000$ $p = \frac{1}{12,000,000}$

- ▶ $\mathbb{P}(X \geq 1) = 1 - (1 - p)^n \approx 0.340759$

- ▶ $\mathbb{P}(X \geq 2) = 1 - (1 - p)^n - np(1 - p)^{n-1} \approx 0.0660758$ 6.6% : not large but not small either

- ▶ $\mathbb{P}(X \geq 2 \mid X \geq 1) = \frac{\mathbb{P}(X \geq 2)}{\mathbb{P}(X \geq 1)} \approx 0.193907$

19.39% : that's pretty high!

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- ▶ n couples

- independently

- ▶ Exam

Upshot: even if the *all of prosecutor's independence assumptions are correct*, the probability they are innocent is not 1 in 12 million, it's more like 20%! That is **reasonable doubt**.

- ▶ $\mathbb{P}(X \geq 1) = 1 - (1 - p)^n \approx 0.151875$

- ▶ $\mathbb{P}(X \geq 2) = 1 - (1 - p)^n - np(1 - p)^{n-1} \approx 0.0660758$ **6.6% : not large but not small either**

- ▶ $\mathbb{P}(X \geq 2 \mid X \geq 1) = \frac{\mathbb{P}(X \geq 2)}{\mathbb{P}(X \geq 1)} \approx 0.193907$

19.39% : that's pretty high!

CORRELATION

Black man with beard	1 in 10
Man with mustache	1 in 4
White woman with pony tail	1 in 10
White woman with blonde hair	1 in 3
Yellow motor car	1 in 10
Interracial couple in car	1 in 1,000

If you are a white woman with blonde hair, you are more likely to be a white woman with a pony tail.

It's possible that if you are blonde, you are more likely to pick a yellow car.

CORRELATION

Conditioned on
events above

Black man with beard	1 in 10
Man with mustache	1 in 2
White woman with pony tail	1 in 10
White woman with blonde hair	1 in 2
Yellow motor car	1 in 5
Interracial couple in car	1 in 1,000

New probability (using the chain rule): 1 in 2 million
instead of 1 in 12 million.

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- ▶ n couples, each matching description with probability p , independently of the other couples

Example: $n = 5,000,000$ $p = \frac{1}{2,000,000}$

$$\mathbb{P}(X \geq 1) = 1 - (1 - p)^n \approx 0.9179 \quad 71.27\% : \text{huge!}$$

$$\mathbb{P}(X \geq 2) = 1 - (1 - p)^n - np(1 - p)^{n-1} \approx 0.7127$$

$$\mathbb{P}(X \geq 2 \mid X \geq 1) = \frac{\mathbb{P}(X \geq 2)}{\mathbb{P}(X \geq 1)} \approx 0.7764$$

77.64% : huge!

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► n co
inde

Upshot: if the prosecutor's independence assumptions are even mildly wrong, the probability they are innocent jumps to **over 75%**. The prosecutor's case falls apart when analyzed with a little more care.

$$\mathbb{P}(X \geq 1) = 1 - (1 - p)^n \approx 0.9179 \quad 71.27\% : \text{huge!}$$

$$\mathbb{P}(X \geq 2) = 1 - (1 - p)^n - np(1 - p)^{n-1} \approx \boxed{0.7127}$$

$$\mathbb{P}(X \geq 2 \mid X \geq 1) = \frac{\mathbb{P}(X \geq 2)}{\mathbb{P}(X \geq 1)} \approx \boxed{0.7764}$$

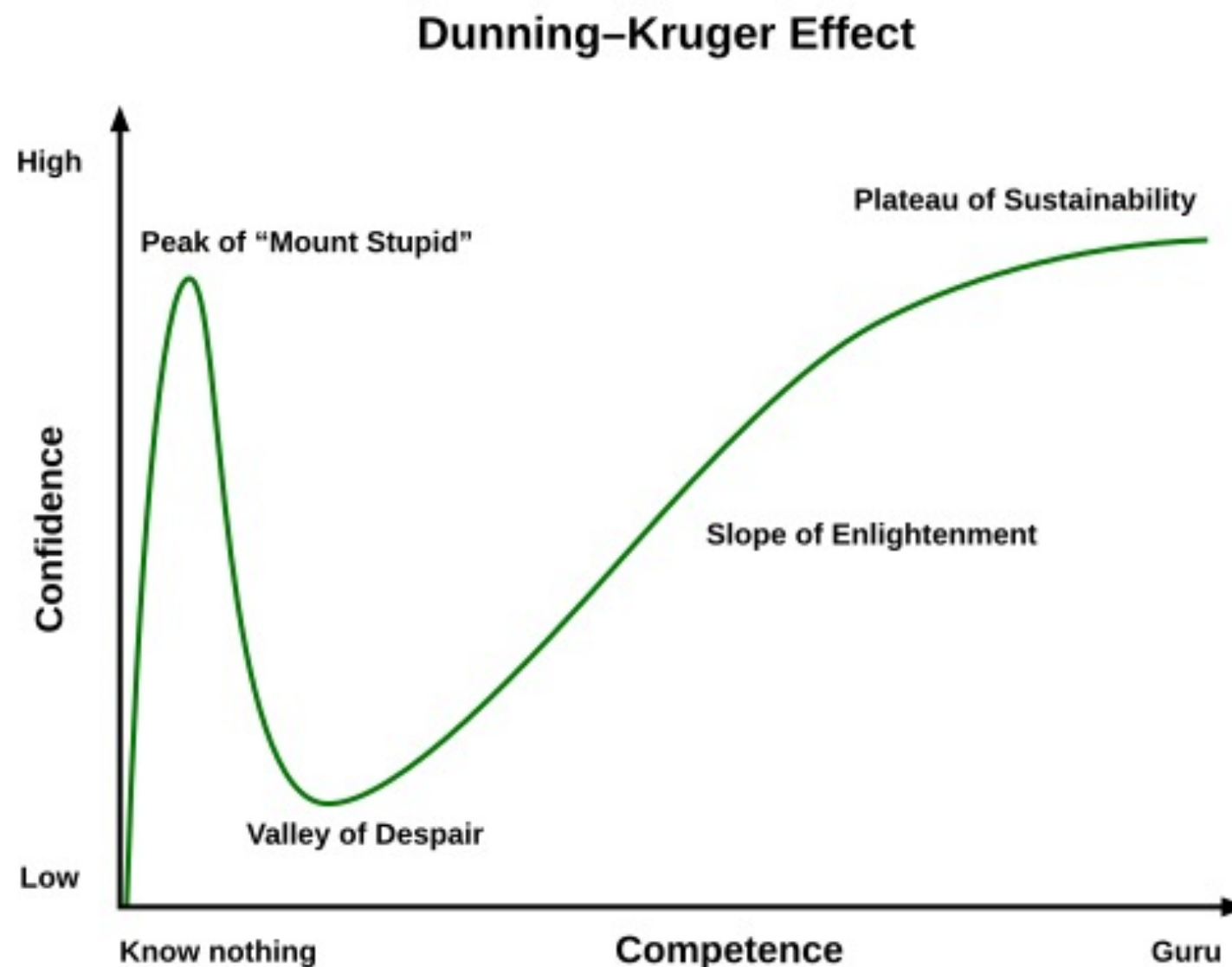
77.64% : huge!

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- ▶ **Lesson 1:** Don't fall prey to the **Dunning-Kruger effect**, where since you know a *little* about something, you overestimate your abilities.

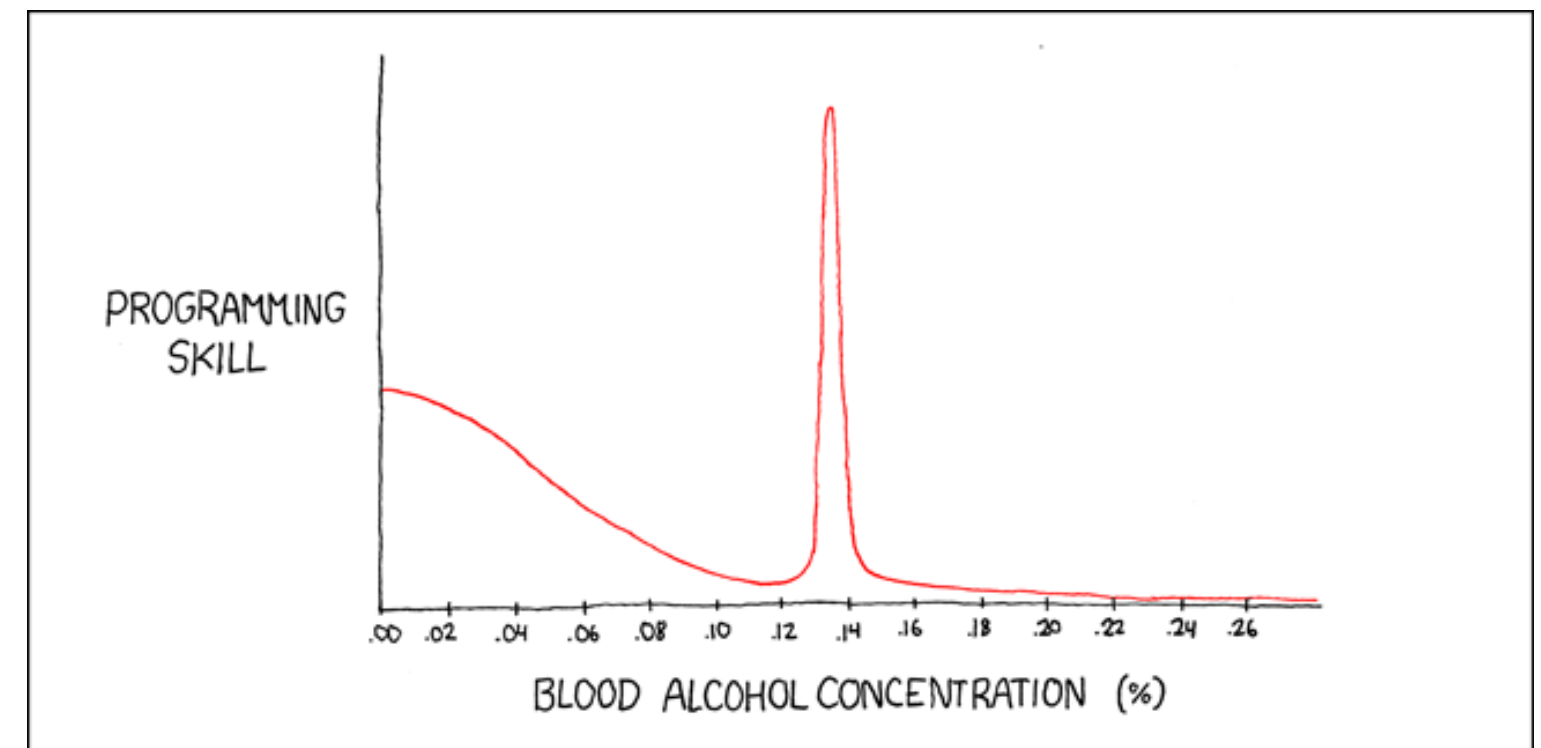
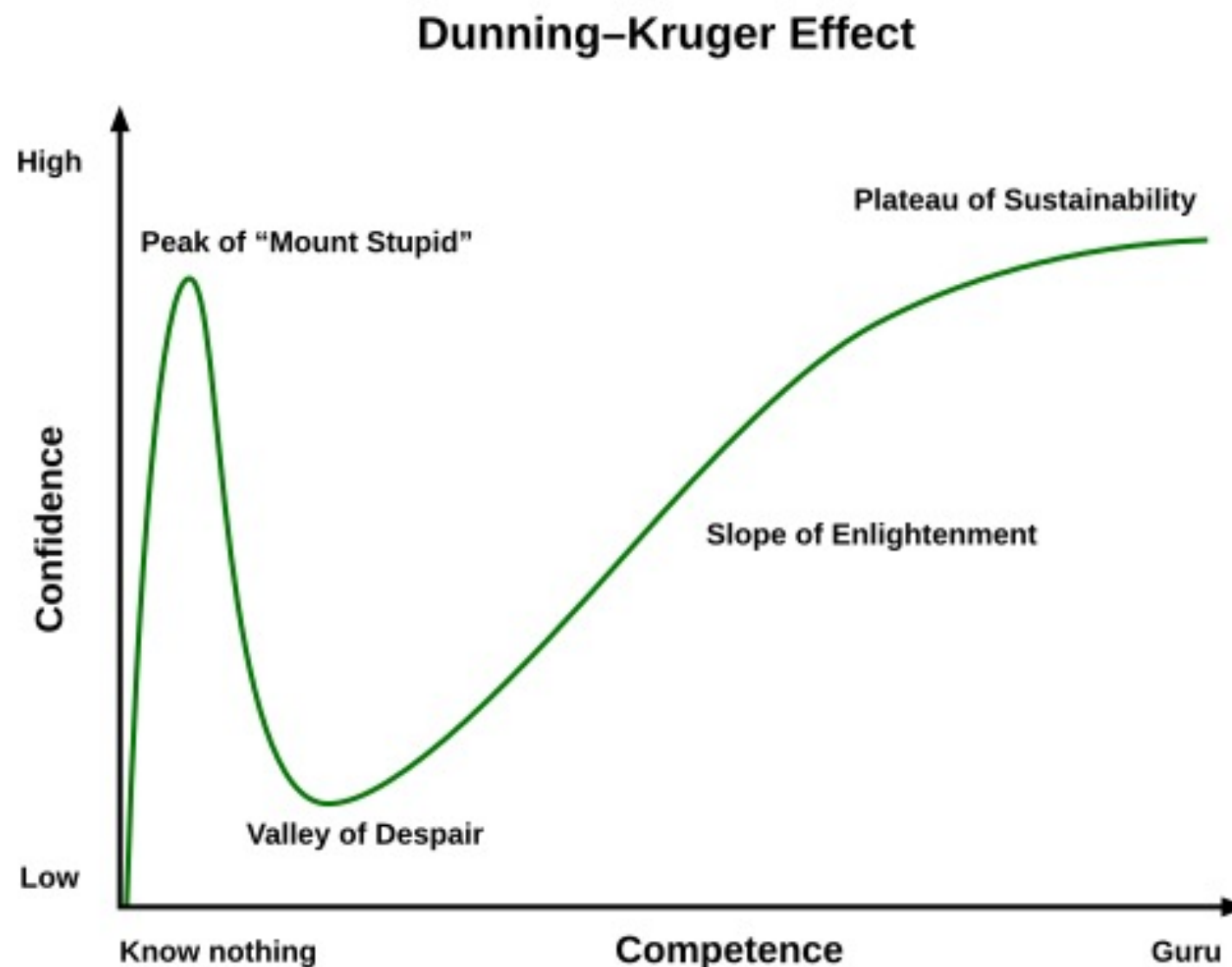
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Not to be confused with
Ballmer's peak

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- ▶ **Lesson 1:** Don't fall prey to the **Dunning-Kruger effect**, where since you know a *little* about something, you overestimate your abilities.
- ▶ The mathematician involved in the case knew the product rule, and overconfidently thought that meant he could model this complex legal situation.

PEOPLE V. COLLINS

- ▶ **Lesson 1:** Don't fall prey to the *Dunning-Kruger effect*, where since you know a *little* about something, you overestimate your abilities.
- ▶ The mathematician involved in the case knew the product rule, and overconfidently thought that meant he could model this complex legal situation.
- ▶ **Lesson 2:** Math is a powerful tool. Don't use it for bad things 😊

PEOPLE V. COLLINS

- ▶ There are at least three flaws in the prosecution's argument:
 - ▶ Independence assumptions: "black man with beard" and "man with mustache" may not be independent
 - ▶ Not taking into account the prosecutor's fallacy (there are a lot of people in the LA region: lots of false positives)
 - ▶ Not conditioning on all the available information ("at least one such couple")

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- ▶ The California Supreme Court overruled the initial guilty verdict

NEXT TIME: INDEPENDENT RANDOM VARIABLES



Hoping to avoid the difficulties of using conditional probability, Thomas Jefferson writes the Declaration of Independence.